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Employment

August 2021–Present	Assistant Professor, Pontificia Universidad Catolica de Chile
July 2020–May 2021	Postdoctoral position, Max Planck Institute for Mathematics in the Sciences, Leipzig, Germany
January 2018– May 2020	Visiting Assistant Professor, University of Kansas
Fall 2017	MSRI Postdoctoral Fellow
2012–2017	Teaching Assistant, University of California, Davis

Education

- Ph.D. in mathematics, University of California, Davis, 2017
- Bachelor in mathematics, Universidad de Los Andes, Colombia 2012

Research Activities

Research Interests Discrete geometry, algebraic combinatorics. Particular topics of interests include matroids, generalized permutohedra, toric varieties, lattice polytopes and Ehrhart theory.

Published articles

- (1) Foundation of one-particle reduced density matrix functional theory for excited states (with Jean-Philippe Labb, Julia Liebert, and Christian Schilling)
Journal of Chemical Theory and Computation, to appear.
- (2) Minkowski summands of cubes (with Joseph Doolittle, Bennet Goeckner, Michael S. Ross, and Li Ying.)
Bulletin of London Mathematical Society, to appear.
- (3) On the Todd class of the Permutohedral variety (with Fu Liu)
Algebraic Combinatorics, **4** (2021).
- (4) When are multidegrees positive? (with Yairon Cid-Ruiz, Binglin Li, Jonathan Montano, and Naizhen Zhang)
Advances in Mathematics, **374** (2020)
- (5) Finiteness theorems for matroid complexes with prescribed topology (with Jose Samper)
European Journal of Combinatorics, **92** (2021)
- (6) Deformation cones of nested braid fans (with Fu Liu)
International Math. Research Notices, to appear
- (7) Coxeter submodular functions and deformations of Coxeter permutahedra (with Federico Ardila, Chris Eur, and Alexander Postnikov)
Advances in Mathematics **365** (2020)

- (8) A pithy look at the polytope algebra
Proceedings of The 2018 Summer Workshop on Lattice Polytopes, Osaka University (2018).
- (9) Smooth polytopes with negative Ehrhart coefficients (with Fu Liu, Benjamin Nill, and Andreas Paffenholz)
Journal of Combinatorial Theory Ser. A **160** (2018), 316-331
- (10) Berline-Vergne valuation and generalized permutohedra (with Fu Liu)
Discrete and Computational Geometry **60** (2018), no. 4, 885-908.
- (11) The topology of the external activity complex of a matroid (with Federico Ardila and Jose Samper)
Electronic Journal of Combinatorics, **23** (2016), P3.8.
- (12) The arithmetic Tutte polynomials of the classical root systems (with Federico Ardila and Mike Henley)
International Math. Research Notices, (2015) 2015 3830-3877.

Preprints

- (1) Partial Permutohedra (with Roger Behrend, Anastasia Chavez, Alexander Diaz-Lopez, Laura Escobar, Pamela E. Harris, and Erik Insko) [arXiv:2207.14253](https://arxiv.org/abs/2207.14253)
- (2) Double Schubert polynomials do have saturated Newton polytopes (with Yairon Cid-Ruiz, Fatemeh Mohammadi, and Jonathan Montaña) [arXiv:2109.10299](https://arxiv.org/abs/2109.10299)
- (3) The permuto-associahedron revisited (with Fu Liu) [arXiv:2109.08658](https://arxiv.org/abs/2109.08658)
- (4) Common tangents to polytopes, (with Joseph Doolittle, and Jose Samper) [arXiv:2108.13569](https://arxiv.org/abs/2108.13569)
- (5) An effective solution to convex 1-body N-representability (with Jean-Philippe Labb, Julia Liebert, Arnau Padrol, Eva Philippe, and Christian Schilling) [arXiv:2105.06459](https://arxiv.org/abs/2105.06459)
- (6) Hopf monoids of ordered simplicial complexes (with Jeremy Martin and Jose Samper) [arXiv:2004.07308](https://arxiv.org/abs/2004.07308)

Awards

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- Alice Leung Scholarship excellence in Mathematics, 2016.
This scholarship is given to the graduate student who has shown excellence in Mathematics, including research, scholarship and teaching.
- Graduate Student Travel Award, UC Davis, 2016.
- Miguel Velez Scholarship for Latin American students UC Davis, 2015.
This award is made on the basis of character, ability and financial need, with preference given to students from Colombia, South America.
- Bronze Medal at the 48th International Math Olympiad.

Invited Talks at Conferences

<i>New Developments in Momentum Polytope Theory</i>	ICMAT, online	July 2021
<i>Algebraic Geometry and Polyhedra</i>	ICERM, online	April 2021

<i>KUMUNU 2019</i>	Lincoln, NE	September 2019
<i>Coloquio Latinoamericano de Algebra</i>	Mexico DF, Mexico	August 2019
<i>Summer Workshop on Lattice Polytopes</i>	Osaka, Japan	August 2018
<i>ALTENCOA 8(Minicourse)</i>	Popayan, Colombia	July 2018
<i>Great Plains Combinatorics Conference</i>	Manhattan, KS	April 2018
<i>Bay Area Discrete Math Day</i>	Oakland, CA	November 2017
<i>Algebraic and Geometric Combinatorics on Convex Polytopes</i>	Osaka, Japan	January 2017
<i>Einstein Workshop on Lattice Polytopes</i>	Berlin, Germany	December 2016

Seminar Talks

<i>Institut fur Mathematik Postdam</i>	Online	July 2022
<i>Goethe-Universitat</i>	Frankfurt	July 2022
<i>University of Copenhagen</i>	Online	July 2022
<i>University of Waterloo</i>	Online	March 2022
<i>University of Wyoming</i>	Online	November 2021
<i>Seminario Sabanero de Combinatoria</i>	Online	October 2021
<i>Seminario Geometria Algebraica (PUC)</i>	Santiago	October 2021
<i>Nonlinear Algebra Seminar, MPI</i>	Leipzig	June 2021
<i>Institut de Mathematiques de Jussieu</i>	Online	May 2021
<i>Ghent University</i>	Online	May 2021
<i>Freie Universitat, Berlin</i>	Online	April 2021
<i>Seminario Pregunton, UNAM</i>	Online	November 2020
<i>KTH Royal Institute of Technology, Stockholm</i>	Online	October 2020
<i>Technische Universitat Berlin</i>	Online	October 2020
<i>Colorado State, Boulder</i>	Online	August 2020
<i>Nonlinear Algebra Seminar, Leipzig</i>	Online	August 2020
<i>University of California, Davis</i>	Online	May 2020
<i>Ohio State University</i>	Columbus	November 2019
<i>New Mexico State University</i>	Las Cruces, NM	November 2019
<i>Freie Universitat</i>	Berlin, Germany	July 2019
<i>University of Miami</i>	Miami, FL	May 2019
<i>University of California, Davis</i>	Davis, CA	March 2019
<i>Washington University</i>	St Louis, MI	November 2018
<i>San Francisco State University</i>	San Francisco	October 2018
<i>Reed's College Math Colloquium</i>	Portland, OR	April 2018
<i>Saint Mary College</i>	Oakland, CA	November 2017
<i>San Jose State University</i>	San Jose, CA	November 2017
<i>University of Washington</i>	Seattle, WA	February 2017
<i>University of California, Los Angeles</i>	Los Angeles, CA	November 2016
<i>San Francisco State University</i>	San Francisco, CA	April 2015

Talks at American Mathematical Society Special Sessions

<i>Algebra and Combinatorics of Polytopes</i>	San Francisco, CA	May 2021
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<i>Algebraic and topological combinatorics</i>	El Paso, TX	September 2020
<i>Geometric and Topological Combinatorics</i>	Gainesville, FL	November 2019
<i>Combinatorial Algebraic Geometry</i>	Madison, WI	September 2019
<i>Algebraic and Geometric Combinatorics</i>	Madison, WI	September 2019
<i>Geometric and Topological Combinatorics</i>	Honolulu, HI	March 2019
<i>Combinatorial Commutative Algebra and Polyhedral Geometry</i>	Hartford, CO	April 2019
<i>Commutative Algebra</i>	Portland, OR	March 2018
<i>Polynomials in Enumerative, Algebraic, and Geometric Combinatorics</i>	Buffalo, NY	September 2017
<i>Geometric Combinatorics and Combinatorial Commutative Algebra</i>	Denton, TX	September 2017
<i>Combinatorial and Computational Commutative Algebra and Algebraic Geometry</i>	Salt Lake City, UT	April 2016
<i>Topological Combinatorics</i>	Seattle, WA	January 2016

Contributed Conference Talks

<i>32nd International Conference on Formal Power Series and Algebraic Combinatorics</i>	Online	July 2020
<i>CombinaTexas</i>	College Station, TX	February 2018
<i>Graduate Student Combinatorics Conference</i>	Lawrence, KS	April 2017
<i>CIMPA Research School: Algebraic, Enumerative and Geometric Combinatorics - ECCO 2016</i>	Medellin, Colombia	June 2016
<i>UC Berkeley/Davis Combinatorics Gathering</i>	Berkeley, CA	October 2014
<i>Encuentro Colombiano de Combinatoria - ECCO 2014</i>	Bogota, Colombia	June 2014
<i>Graduate Student Combinatorics Conference</i>	Minneapolis, MN	April 2013
<i>Encuentro Colombiano de Combinatoria - ECCO 2012</i>	Bogota, Colombia	June 2012
<i>Congreso Colombiano de Matematicas</i>	Bucaramanga, Colombia	July 2011

Poster Presentations

<i>32nd International Conference on Formal Power Series and Algebraic Combinatorics</i>	Online	July 2020
<i>31st International Conference on Formal Power Series and Algebraic Combinatorics</i>	Ljubljana, Slovenia	July 2019
<i>Introductory Workshop, Combinatorial Algebraic Geometry thematic program</i>	Toronto, Canada	August 2016
<i>28th International Conference on Formal Power Series and Algebraic Combinatorics</i>	Vancouver, Canada	July 2016
<i>2nd Algorithmic and Enumerative Combinatorics - Summer School</i>	Hagenberg, Austria	July 2015

<i>27th International Conference on Formal Power Series and Algebraic Combinatorics</i>	Daejeon, Korea	June 2015
<i>26th International Conference on Formal Power Series and Algebraic Combinatorics</i>	Chicago, IL	June 2014

Teaching and Advising Activities

Courses at University of Kansas

- Math 220, Differential Equations (S'19)
- Math 290, Elementary Linear Algebra (S'18, F'18)
- Math 409, Topics in Geometry for Secondary/Middle School Teachers (S'19)
- Math 410, Topics in History of Math for Secondary/Middle School Teachers (S'19)
- Math 724, Combinatorial Mathematics (F'19)
- Math 725, Graph Theory (S'18)

Courses at University of California, Davis

- Math 16A, Short Calculus (F'15)
- Math 16B, Short Calculus (F'16)
- Math 16C, Short Calculus (Summer '15)

Professional service and outreach

- **Referee** Journal of Combinatorial Theory Series A, Discrete & Computational Geometry, SIAM Journal on Discrete Mathematics, Discrete Mathematics, International Research Mathematical Notices, Electronic Journal of Combinatorics, Selecta Mathematica.
- **Co-organizer** (Polytop)ics: Recent advances on polytopes April 6-9, 2021.
- **Co-organizer** Combinatorics Seminar at KU (2018-2020).
- **Member of the Organizing Committee** Combinatorics meets Algebra, Geometry and Optimization ECCO, Universidad del Norte, Barranquilla, Colombia. (June 2018)
- **Speaker** at Nerd Nite Lawrence. (May 2019 and November 2019)
- **Co-organizer** AMS Special Session on Multiplicities and Volumes: An Interplay Among Algebra, Combinatorics, and Geometry, Ann Arbor, MI. (October 2018)
- **Co-organizer** of UC Berkeley/Davis Combinatorics Gathering 2015-2016.
- **Organizer** of the Student Run Discrete Math seminar, UC Davis 2013-2014.
- **Organizing committee** of Colombian National Math Olympiad 2008-2012.